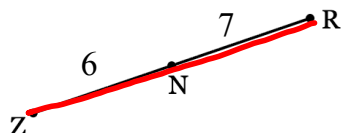


Students can measure and compare segments and use the segment addition postulate to find missing lengths.

Do Now

- 1) Place your name tent on your desk.
- 2) In the diagram below, points Z , N , and R are collinear.



$\overline{ZR}$

- a. Give another name for $\overline{ZR}$.
- b. Make a prediction: Yesterday, we used $\overline{ZN}$ to name a geometric figure. What do you think $ZN = 6$ means?

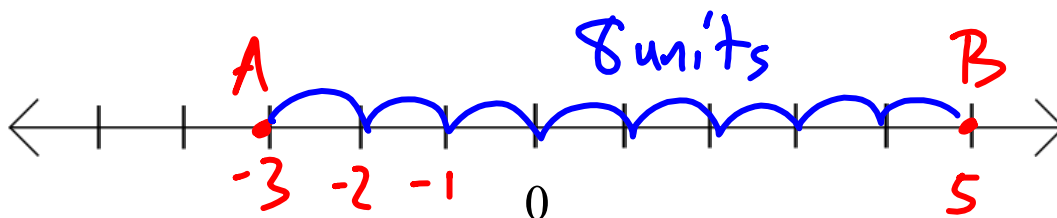
$\overline{ZN}$ is 6 units long

- c. Make a prediction: " $ZR = \underline{13}$ ".

$$6 + 7 = 13$$

Do Next

- 3) Plot and label points $A = -3$ and $B = 5$ on the number line below.



- 4) Find AB .

8

~~AB = 6~~

Segments & Their Lengths

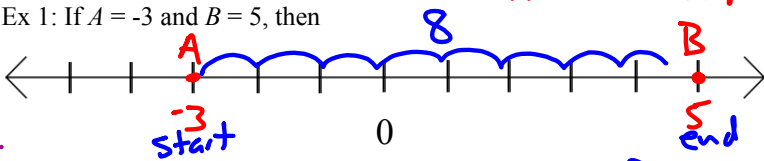
Important Notation:

$\overline{AB}$ means a geometric figure.

AB means the length of the geometric figure $\overline{AB}$. This is a #.

To determine AB , the length of segment $\overline{AB}$, we would need to place $\overline{AB}$ over a number line and find the absolute value of the difference between its start and end points. *How many units in between endpoints*

Ex 1: If $A = -3$ and $B = 5$, then



$$AB = |(-3) - (5)| = |-3 - 5| = |-8| = 8$$

Consider: why can't the length of a segment be negative?

Congruent Segments

The word congruent ($\cong$) in geometry means equal in every possible way. In fact, it means that we can lift one geometric object and move it so that it fits exactly on top of the other, a perfect overlap.

Line segments that have the same length are said to be congruent.

Lengths are equal

 $AB = CD$

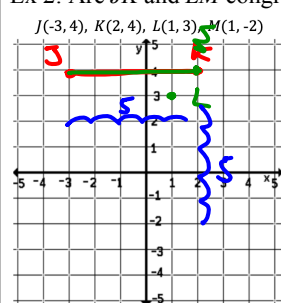
$\iff$

"iff"

Segments congruent

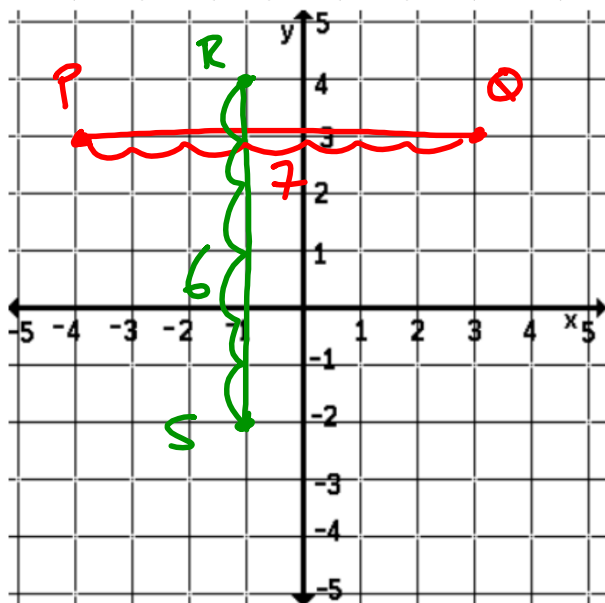
 $\overline{AB} \cong \overline{CD}$

Ex 2: Are $\overline{JK}$ and $\overline{LM}$ congruent? Justify your answer.



Yes, $\overline{JK} \cong \overline{LM}$ because they have the same length.

You Try:

1) Are $\overline{PQ}$ and $\overline{RS}$ congruent? Justify your answer. $P(-4, 3)$, $Q(3, 3)$, $R(-1, 4)$, $S(-1, -2)$ 

$$\overline{PQ} \neq \overline{RS}$$

because

$$PQ = 7$$

and

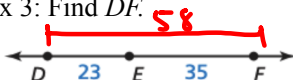
$$RS = 6$$

Segment Addition PostulateLet B be collinear with A and C so that it is somewhere between them.

— something that must be true.

The Segment Addition Postulate tells us that

$$\overline{AB} + \overline{BC} = \overline{AC}$$

In other words, "the whole is equal to the sum of its parts."Ex 3: Find DF .

$$DF = 23 + 35 = \boxed{58}$$

Ex 4: Find GH .

$$\begin{array}{r} 21 + GH = 36 \\ -21 \quad -21 \end{array}$$

$$\boxed{GH = 15}$$